

Parul Institute of Engineering & Technology

Electronics & Communication Engineering Department

ASSIGNMENT-2

Subject Name: DIGITAL ELECTRONICS

Subject Code: 303105220

(Chapter-2- Minimization Techniques)

1. Given Boolean function
 $F = x y + x' y' + y' z$
 - a. Implement it with only OR & NOT gates
 - b. Implement it with only AND & NOT gates
2. Express following Function in Product of Maxterms
 $F(x,y,z) = (xy + z)(y + xz)$
3. Explain briefly : SOP & POS , minterm & maxterm , canonical form , (05)
propagation delay, fan out
4. What is the principle of Duality Theorem?
5. Explain briefly: standard SOP and POS forms.
6. What are Minterms and Maxterms?
7. Define: Noise margin , Propagation delay
8. Reduce the expression:
 - a. $A+B(AC+(B+C')D)$
 - b. $(A+(BC)')'(AB'+ABC)$
9. Define : Integrated Circuit and briefly explain SSI, MSI, LSI and VLSI
10. Draw the logic symbol and construct the truth table for each of the .
 - [1] Two input NAND gate
 - [2] Three input OR gate
 - [3] Three input EX-NOR gate
 - [4] NOT gate
11. Give classification of Logic Families and compare CMOS and TTL Families
12. Demonstrate by means of truth tables the validity of the following Theorems of Boolean algebra
 - (i) De Morgan's theorems for three variables
 - (ii) The Distributive law of + over-
13. Express the following functions in sum of min terms and product of max terms:
 - a. $F(A,B,C,D) = D(A'+B) + B'D$
 - b. $F(A,B,C) = (A'+B)(B'+C)$
 - c. $F(x,y,z) = 1$